

a real proof released a payment that never happened

The proof is real. The payment is wrong.

The bench, the drop-in library, the settlement rail, and the check on value entering the chain, all closing the third check a valid Attestcoin proof leaves open.

15/16

protocol entry points exercised. A typical consumer touches one.

48/53

submissions miss at least one binding check, measured across the field.

11/11

claims reproduce off the public chain, no key, no clone.

01 A valid proof is not permission to pay

The BlockProver precompile at **0x0FD2** proves inclusion and continuity, then stops. Everything that decides whether money should move is left to the integrator.

01

Transaction status

02

Emitting contract

03

Event signature

04

Replay guard

05

Chain identity

06

Order and field binding

07

Block window

08

Log selection

5 of 53

We scored every submission in this hackathon against the twelve checks. Only five clear all of them. 91% ship at least one gap. Findings go to the affected teams and Creditcoin in private, first.

02 Four moving parts, one guarantee

A falsifier

A bench that releases a real deployed escrow against a legitimate proof of the wrong thing, next to a hardened one that rejects. On-chain, side by side.

A drop-in library

ThirdCheckLib implements the correct path in two calls, `verifyReceipt` and `bindPayment`, so a consumer cannot reimplement the checks and get one wrong.

A settlement rail

SettlementHub runs the full third check by construction for every order; VerifiedRegistry lets a verified operator settle at a lower fee. The fee on safe settlement is the model.

A CI gate

A static analyzer that fails a build before mainnet if a consumer skips the third check, the way a linter or a test suite would.

Depth is the point: ThirdCheck exercises **15 of 16** protocol entry points across BlockProver and ChainInfo, plus the full 16-method EvmV1Decoder surface, where the official examples touch a handful.

03 One order, three parties, two chains



seller received

+0.0009975

protocol fee · 0.25%

+0.0000025

distinct addresses

3

Any lending, RWA or trading app routes orders through the hub and settles safe by construction. **The fee on safe settlement is the revenue; audit-grade review is the service.**

```
$ npm run judge:verify · or verify on the live site, no clone. tampered proof fails, real one reproduces 11/11
```

04 Verified Inflows: the check on money coming in

The costliest step in cross-chain is crediting an inbound deposit. Roughly **\$2B** in bridge losses came from crediting a transfer without independently confirming it. The same third check, aimed inbound.



fresh beneficiary

bridge trusted

0 → 0.001 **none**

Value and users cross into Creditcoin from larger chains only when the crossing is safe. **This is the check that makes it safe, and the reason liquidity can come in.**

`bindDeposit` is a second predicate over the same verified receipt · the library generalises past payments without loosening a check

05 Live contracts and where this goes

CONTRACT	NETWORK	ADDRESS
SourceSettlement	Ethereum Sepolia	0xD8504B263104aa915974eCCE1002d7F7587e88cD
SourceGateway	Ethereum Sepolia	0x879628662310232F9c287eF45d14d89B7cD5886E
HardenedEscrow	Creditcoin CC3	0xeC82270dc356948FC2e5E969a887ce6bE27e375A
VerifiedRegistry	Creditcoin CC3	0xa2E744fEa8707aE124ee7d605D2fc1b58BF68752
SettlementHub	Creditcoin CC3	0x676a74fa6542BEd2dD4A16EF122f75968329B1B0
InflowConsumer	Creditcoin CC3	0x037D8E868Ced6F3612EfEd070aBB316eF8fB82c0
CreditLineApp	Creditcoin CC3	0xE6049333594A454E5A73C38a8a3DaAC8D90191f7

A reference app already builds on the library: a cross-chain credit line that opens from verified inbound collateral. Next: the default pre-mainnet gate for every Attestcoin consumer. Continuous verification in CI, runtime monitoring of deployed consumers, and audit-grade review as a service, while the gate and the library stay free and drive adoption.